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# Identification and Classification of Some Common Wild Edible Mushrooms Collected and Consumed by the Tribes of Jharkhand

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**Abstract :**As Jharkhand is thickly populated with different tribes, its ethno-mycological knowledge is rich. During the rainy season especially in the beginning of it varieties of wild edible as well as poisonous mushrooms appear in the forest of Jharkhand. The tribal people over here have traditional knowledge which is based on their experiences. They collect many wild edible mushrooms for their consumption or for sale and earn quite a good sum of money because these mushrooms are sold with high price in the village market as well as in the city. The tribal women play a vital role in collecting these mushrooms since they have to look for nutritional food for their children. This paper is an effort to identify these wild edible mushrooms, their uses, scientific names and classifications so that they could be known to others and could be used for further research.

**Keywords: Wild, Edible mushrooms, The tribal people, Identification, Classification.**

## INTRODUCTION

The mycologists have used various characteristics for mushrooms identification and classification like morphology, occurrence, natural habitat, spore born in ascus or basidium, spore print, structure and texture of fruiting bodies etc but the identification of their edibility is based on trial and error method. The practice of eating mushroom probably began during the hunting and gathering period in human prehistory but human learnt it out of his/her experience. The best way to learn mushrooms is to have a local expert teach you[1]. If you are new to identifying and gathering mushrooms, going out to the field with an experienced harvester is recommended[2]. Malaysian wild edible mushrooms, *Termitomyces, Calvatia,* have been successfully documented for further updating and scientific identification, including their genetic sequences in the near future[3]. In context to Nepal, ethno-mycological studies have been conducted by various researchers[4]. In Jharkhand; the tribal people are good ethno-mycologists since they have interacted with fungi for generations. In this article the edibility of naturally growing mushrooms has been identified with the help of the tribal people and their scientific names and taxonomic positions have been found out using mushroom guides and identifying keys. The purpose of this research is to make it known to the public that these mushrooms are edible. They could be domesticated and cultured in a large scale. The traditional knowledge needs to marry technology, for further research.

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## MATERIALS AND METHOD

The field study of some of the districts of Jharkhand namely, Ranchi, Khunti and Simdega was done during rainy season of the years 2012 and 2013 with some tribal women as they have good knowledge of wild edible mushrooms and the places of their occurrences. The natural habitats of these mushrooms were closely studied, their photographs were taken and they were collected and brought to lab. The village markets as well as city markets were also visited to find out what were the different wild

edible mushrooms being sold with what price. Their spore prints were taken and studied. Mushrooms collected were identified with the help of some expert mycologists and mushroom keys and guides, the taxonomic positions were found out. Their nutritional and medicinal uses were found from different sources.

## RESULTS AND DISCUSSION

Following are some of the identified mushrooms



**Fig. 1. *Macrolepiota procera* (Scop.) Singer**

*Class: Basidiomycetes, Order: Agaricales (Hymenomycetes), Family: Agaricaceae, Genus: Macrolepiota , Species: procera* (Scop.) Singer

Spores are formed in basidium, basidia borne on hymenium which is spread over gills therefore it is placed under gill mushrooms. It has a tall stipe (up to 25 cm) and a large cap with rusty brown coloured umbo. It is coated with pink granules, spore print is white. June is the best month for this mushroom. It is usually found on the heap of compost manure. Rain always precedes the emergence of this mushroom. It tastes good.

**Nutritional value:** it contains proteins, iron, zinc, chitin, chitosan, fiber, vitamins and minerals, glycerol, mannitol, glucose, trehalose, lepiotan and about 20 amino acids[5]. Its protein content is 32% on dry weight basis[6]. Nutritional value of *Macrolepiota* is moisture 91.0%, ash 1.09%, protein 3.30%, fat 0.18%, and carbohydrate 0.86% 7.

**Medicinal uses**: it is used as vegetable in food. It has anti-tumor activity, exhibit an antibiotic activity against gram negative bacteria.



***Termitomyces clypeatus R.* Heim**

*Phylum: Basidiomycota, Class: Hymenomycetes, Order: Agaricales, Family: Tricholomataceae, Genus: Termitomyces, species: T. clypeatus R.* Heim

The distinguishing character of this mushroom is the prominent, cone -like pointed, dark colour projection at the centre of the cap. It is found between June and August. It grows single and not found abundantly. This is one of the most delicious mushrooms found in nature[8]. There is one other mushroom which looks similar to it, is poisonous.



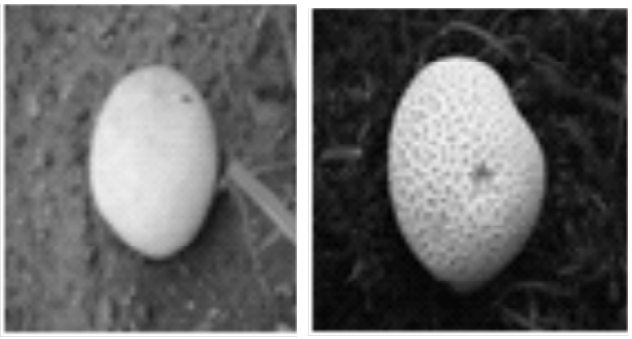
***T. albuminosa* (Berk.) R.**

In Jharkhand this mushroom is known as *badakhukhri* (big mushroom). It comes up exclusively from termite's nests[9]. A patch of veil is present at the top of the pileus. The skin of the cap peels off. The stalk is about 25 cm long and swollen near the attachment with the cap. The spore print is deep pink. Found between July and August. It is a very delicious mushroom. There is another species, *T. heimii* K. Natarajan which looks similar to *Termitomyces albiminosa* but little smaller in size and appears at least one month before *Termitomyces albiminosa* appears which has annulus and is most abundantly found mushroom and locally known as *patiyari khukhri*. *T. microcarpous* (Var) *is* another species of *Termitomyces*

which is very small in size and grows in clumps especially on the sandy soil; therefore it is called as, *Balukhukhri (sand mushroom).*

**Nutritional value:** *Termitomyces* is an important source of enzymes of industrial importance such as xylanase, amylase, and cellulase, and is rich source of sugar, protein fibre, lipid, and vitamin, mineral which is used in lowering blood pressure, rheumatism, kwashiorkor, obesity, diarrhoea and as purgative[10]. Its nutritional value moisture is 91.3%, ash 0.81%, protein 4.1%, fat 0.22% and carbohydrate 1.13%[11]. *Termitomyces microcarpus* has moisture content 90.67±0.35 dry matter 9.33±0.35 carbohydrate 46.53±1.01 lipid 2.33±0.32 protein 29.4±1.25 fibre 11.5±0.72 ash 11.2±0.56. *Termitomyces heimii* has moisture content 92.5±0.44 dry matter 7.5±0.44 carbohydrates 39.03±0.96 lipids 2.11±0.4 protein 34.2±1 fibre 9.73±0.42 ash 16.8±0.4 [12].

**Medicinal uses:** it contains antioxidant compounds such as polyphenol and vitamin C, protein 31.4–36.4% and immune stimulatory agent[13]. The extract exhibits antioxidant activities lower blood pressure in hypertensive patients. It is also used for cholesterol lowering and immunostimulatory effects, malnutrition, weakness and other nutritional disorder, Rheumatism, diarrhoea, lowering high blood, blood tonic.



***Lycoperdon and Calvatia***

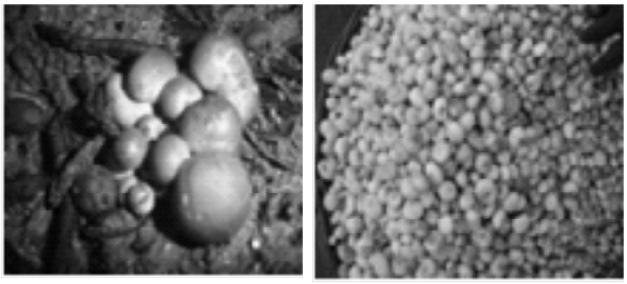
*Class: Basidiomycetes, Order: Lycoperdales (Gasteromycetes) (Stomach fungi)*

*Family: Lycoperdaceae, Sub-family: Lycoperdae, Genus: Lycoperdon, Calvatia*

Spores are formed in basidium; basidia are borne within a container (peridium). The order Lycoperdales has been divided into four or five families, like, Lycoperdaceae, Geastraceae etc. *Lycoperdon* is a genus of puffball mushrooms. Fruit body is spherical or pear shaped with round top and tapering stem, 1-6 cm wide, 3-7 cm high. It grows in the beginning of the rainy season. In ancient times, the spore mass of ripe puffballs were used to stop bleeding which was probably due to the good virtue of its powdery nature[14]. *Lycoperdon pyriforme, L. gemmatum and L. umbrinum* are common species. *Lycoperdon* grows fast and decay very soon. There seems to be no pore at the centre of the fruiting body for spore dispersal, fruiting body decays and the spores are released. They should be consumed when they are very young and spores are not matured.

**Nutritional value***: L. giganteum* contains moisture 95.7d dry matter 4.3g soluble total glycogen 17.5a 10.0bc 25.8a 23.3e 5.57g 15.5b [15].

**Medicinal uses**: Used as vegetable in food. The spores are used as powder to heal wounds and to cure baby rashes and stop bleeding in a fresh wound [16].



***Geastrum***

*Class: Basidiomycetes, Order: Lycoperdales (Gasteromycetes) (Stomach fungi)*

*Family: Lycoperdaceae, Sub-family: Geastrae, Genus: Geaster.*

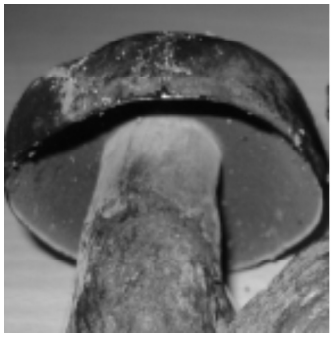
Spores develop as a dry power. Peridia on short stem, peridium conspicuous, breaks away in flakes and becomes star shaped. It is commonly known as earthstars. The name comes from, '*geo'* meaning earth and '*aster'* meaning star. The name refers to the behaviour of the outer peridium. At maturity, the outer layer of the fruiting body splits into segments which turn outward creating a star-like pattern on the ground. The inner peridium is a spore sack. In some species, the outer peridium splits from a middle layer, causing the spore sack to arch off the ground. Outer peridium opens when wet and closes when dry, it is hygroscopic. It is found in June and July months abundantly in the *Sal* forest (*Shorea robusta*). In Jharkhand

local people call it *rugra* or *putu* and desire it equal to meat. Two varieties of *Geaster* are found in Jharkhand.

**Nutritional value**: it contains a number of bioactive compounds, including fungal sterols such as ergosta-4, 6, 8, (14), 22-tetraen-3-one, 5, 6-dihydroergosterol, ergosterol, and peroxyergosterol. The fungus also contains various fatty acids, notably myristic, palmitic, stearic, oleic, alpha-linolenic, and linoeic acid[17].

**Medicinal uses:** The Cherokee put fruit bodies on the navels of babies after childbirth until the withered umbilical cord fell off, "both as a prophylactic and a therapeutic measure"[18]. In traditional Chinese medicine, *G. triplex* is used to reduce inflammation in the respiratory tract, and to staunch bleeding and reduce swelling [19].

***Boletus edulis*** Bull.



*Kingdom: Fungi, Division: Basidiomycota, Class: Agaricomycetes, Order: Boletales, Family: Boletaceae, Genus: Boletus, Species: B. edulis* Bull.

Spores are formed in basidium, basidia are borne on hymenium. Hymenium linings are present inside the tubes therefore they are called tube fungi. Tubes readily separate from the cap and from each other. The name is derived from the Latin term *Bôlçtus* 'mushroom' from the Ancient Greek ultimately from *bôlos* 'lump' or 'clod'. The genus *Boletus* contains many members which are edible and tasty. This is a symbiotic Fungus. The edible *Boletus* of Jharkhand is always found in association with *Syzygium cuminii,* therefore it is called *Jamun Khukhri.* In size, this is the biggest mushroom of Jharkhand. It taste good and appears in the beginning of rainy season.

**Nutritional value:** *Boletus edulis* contains vitamins, minerals and dietary fibre. Fresh mushrooms contain over 80% moisture. Carbohydrates make up the bulk of the fruit bodies, comprising 9.23% of the fresh weight and 65.4% of the dry weight. The carbohydrate component contains the Monosaccharides Glucose, Mannitol and á, á-Trehalose, the Polysaccharide Glycogen, and the water-insoluble structural polysaccharide Chitin, which accounts for up to 80–90% of dry matter in mushroom Cell walls. Chitin, Hemicellulose, and Pectin-like carbohydrates—all indigestible by humans—contribute to the nutritionally desirable high proportion of insoluble fibre in *B. edulis.*

The total Lipid, or crude fat, content makes up 2.6% of the dry matter of the mushroom. The proportions of Fatty acids are: Palmitic acid, 9.8%; Stearic acid, 2.7%; Oleic acid, 36.1%; Linoleic acid, 42.2%, and linolenic acid, 0.2%.

**Medicinal uses**: Used as vegetable in food and to inhibit tumor growth. A lectin isolated from the mycorrhizal mushroom *Boletus edulis* has anticancer activity[20]. It is also used to improve vision and counteract opthalmia and night blindness. It reduces symptoms of essential and renal hypertension and of neurasthenia. Prevent respiratory and digestive tract conditions. Increases blood flow to the brain and heart without increasing blood pressure, decreases heart rate, reduces peripheral and coronary vascular resistance.

Besides the above described mushrooms, *Russula sanguinea* (Bull.)Fr, *Amanita, Clitocybe, Armillaria mellea* (Vahl) P are also found in Jharkhand. *Russula* is found in red, white and yellow colour. White colour one is edible. Millipede is found associated with this mushroom most of the times. It is called *patra* in local language because it is found abundantly with decaying leaves.

Jharkhand has rich diversity of mushrooms. It has deciduous forest, *Shorea robusta* (*Sal* tree) is the main tree over here. Trees shed their leaves which decay and become very good manure for mushrooms. Several species of naturally occurring edible species are being collected and used as vegetable. In the year 2012 in the month of July *rugra* (*Geastrum*) and *Termitomyces* were sold for 400 rupees per kg in Simdega district of Jharkhand. Jharkhand is thickly populated by different tribal groups (*Oraon, Munda, Kharia, Ho, Santhal* etc) who have very good knowledge of wild edible mushrooms. Though the tribal people do not know the nutrient content of these wild edible mushrooms but they consider them equal to

meat and use them as vegetable in food.

Mushrooms have been recognised as the best loved vegetarian food, rich in nutrition, particularly protein.[21] In similar reports indicated the edibility of *Termitomyces* (Ascomycetes) and *Calvatia* (Basidiomycetes) from Malaysia.[22] The morphological characteristics and natural occurrence of some of the wild edible mushrooms of Jharkhand have been identified and classified. Many wild mushrooms are safe to eat, a bit with some caution. Conrad indicated the necessity of a local expert to single out the poisonous ones from the edible ones. As per him, some members of Amanita, an otherwise poisonous genus, are edible. On the other hand, *Chlorophyllum* was regarded as highly poisonous, although it looked like an ordinarily fair and white edible mushroom.[23] I hope this finding will open the door for other researchers to do further study on these mushrooms.

## CONCLUSION

As mushrooms are rich protein food but many of them still remaining wild. The tribal people could help us to identify them since they have good knowledge of wild edible mushrooms as they collect them regularly during rainy season for their consumption. These wild mushrooms need to be tapped as food source by domesticating them; otherwise our knowledge of them will remain hidden in the forest before they are eclipsed by the throes of modernisation.

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